A frozen split of the DGSEM Euler equations with the pressure gradient in product form

MultiRate notes

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Abstract

For split-explicit (MIS) time integration with a *frozen* right-hand side, $N(q) = F(q, q)$, the first argument y is frozen at the stage values and the second z evolves in the fast loop. We describe a DGSEM discretization of the compressible Euler equations in the variables (ρ, $m = \rho\mathbf{u}$, $\Theta = \rho\theta$) in which this split is natural: the pressure gradient is written in product form $\nabla p = P'(\Theta)\nabla\Theta$, the potential temperature θ and the derivative $P' = \mathrm{d}p/\mathrm{d}\Theta$ are frozen exactly where they appear, and the interface flux is a LMARS variant written directly in the momentum m. The fast part is then *linear* in z with coefficients frozen at y, and $F(q, q) = N(q)$ holds by construction.

1 Equations

With $\Theta = \rho\theta$ the pressure is a function of Θ alone,

$$p = P(\Theta) = p_0\Big(\frac{R\,\Theta}{p_0}\Big)^{\gamma}, \qquad P'(\Theta) = \frac{\gamma p}{\Theta} = \frac{c^2}{\theta}, \qquad c^2 = \frac{\gamma p}{\rho}.$$

The Euler equations with gravity, the pressure gradient in product form:

$$\partial_t\rho + \nabla\cdot m = 0, \tag{1}$$

$$\partial_t\Theta + \nabla\cdot(\theta\, m) = 0, \tag{2}$$

$$\partial_t m + \nabla\cdot(m\otimes\mathbf{u}) + P'(\Theta)\,\nabla\Theta + \rho g\,\mathbf{e}_z = 0. \tag{3}$$

Mass and Θ are in conservation form; the momentum equation is not (the pressure term is a non-conservative product), which is of no concern with gravity present. The equations are discretized as they stand, without a reference state and without subtracting a hydrostatic background (Section 3.4).

2 The frozen form

Freeze θ and P' at y, everything else from z:

$$\begin{aligned} F_\rho(y,z) &= -\nabla\cdot m_z, \\ F_\Theta(y,z) &= -\nabla\cdot(\theta_y\, m_z), \\ F_m(y,z) &= \underbrace{-\nabla\cdot(m_y\otimes\mathbf{u}_y)}_{\text{slow } S(y)} \underbrace{-\,P'(\Theta_y)\nabla\Theta_z - \rho_z g\,\mathbf{e}_z}_{\text{fast}}. \end{aligned}$$

$F(q, q)$ is (1)–(3). The fast part

$$F(y,z) - S(y) = L(y)\,z$$

is linear in $z = (\rho, m, \Theta)_z$; its operator $L(y)$ has the coefficients θ_y and $P'(\Theta_y)$ only. Nothing else has to be frozen: the momentum enters linearly, the pressure only through $\nabla\Theta$. A stage of MIS integrates

$$\frac{\mathrm{d}Z}{\mathrm{d}\tau} = \sum_j \frac{\beta_{ij}}{d_i}\big(S(Y_j) + L(Y_j)Z\big) + \sum_j \frac{\gamma_{ij}}{d_i \Delta t}(Y_j - y^n),$$

i.e. with the weighted coefficients $\bar\theta = \sum_j b_{ij}\theta(Y_j)$, $\bar P' = \sum_j b_{ij} P'(\Theta(Y_j))$, $b_{ij} = \beta_{ij}/d_i$ (since L is linear in the coefficients), one linear operator per stage.

3 DGSEM discretization

3.1 Notation

Quadrilateral elements mapped from $[-1,1]^2$, $(\xi,\eta) \mapsto \mathbf{x}$, Jacobian J, metric terms $J\xi_x, J\xi_z, J\eta_x, J\eta_z$ (discretely satisfying the metric identities), tensor-product LGL nodes of degree p, collocated quadrature with weights w_i, differentiation matrix D (SBP: $WD + D^T W = B = \mathrm{diag}(-1, 0, \ldots, 0, 1)$). Contravariant components of a vector v: $\tilde v^\xi = J\xi_x v_x + J\xi_z v_z$, $\tilde v^\eta = J\eta_x v_x + J\eta_z v_z$. At a face node with outward unit normal $\mathbf{n}$ and surface element s (so that $s\,\mathbf{n}$ is the contravariant normal), $v_n = \mathbf{v}\cdot\mathbf{n}$; L/R denote the inner and the outer trace, $\{\{a\}\} = \frac{1}{2}(a_L + a_R)$, $[\![a]\!] = a_R - a_L$.

3.2 Volume terms

In flux-differencing form along the ξ lines (the η lines likewise), node $a = (i,j)$, partner $b = (k,j)$, with the two-point contravariant normal $\tilde n_{ab} = \frac{1}{2}\big((J\xi_x)_a + (J\xi_x)_b,\ (J\xi_z)_a + (J\xi_z)_b\big)$:

$$\begin{aligned}
(\text{mass})_a &= \frac{1}{J_a}\sum_k 2D_{ik}\ \{\{m_z\}\}_{ab}\cdot\tilde n_{ab},\\
(\Theta)_a &= \frac{1}{J_a}\sum_k 2D_{ik}\ \{\{\theta_y\}\}_{ab}\,\{\{m_z\}\}_{ab}\cdot\tilde n_{ab},\\
(\text{advection, slow})_a &= \frac{1}{J_a}\sum_k 2D_{ik}\ \big(\{\{m_y\}\}_{ab}\cdot\tilde n_{ab}\big)\{\{\mathbf{u}_y\}\}_{ab},\\
(\text{pressure})_a &= P'(\Theta_{y,a})\ \frac{1}{J_a}\sum_k D_{ik}\,\Theta_{z,b}\,(J\xi_x, J\xi_z)_b \quad \big(\text{the conservative metric form of } \nabla\Theta\big),\\
(\text{gravity})_a &= \rho_{z,a}\, g\, \mathbf{e}_z.
\end{aligned}$$

Since $\sum_k D_{ik} = 0$ the mass term is the strong-form divergence of m_z; for the Θ term the two-point mean $\{\{\theta_y\}\}$ is the only frozen coefficient. The pressure term is the nodal product of the frozen P'_a with the DG gradient of Θ_z (its surface part in the next section).

3.3 Interface flux: LMARS in the momentum

At a face the normal acoustics of (2)–(3), frozen at a face state (θ_f, c_f), is the linear system

$$\partial_t\Theta + \theta_f\,\partial_n m_n = 0, \qquad \partial_t m_n + \frac{c_f^2}{\theta_f}\,\partial_n\Theta = 0,$$

with characteristic speeds $\pm c_f$ and the Riemann invariants $m_n \pm (c_f/\theta_f)\Theta$. Its exact Riemann solution at the face is

$$m^* = \{\{m_n\}\} - \frac{c_f}{2\theta_f}\,[\![\Theta]\!], \qquad \Theta^* = \{\{\Theta\}\} - \frac{\theta_f}{2c_f}\,[\![m_n]\!]. \tag{4}$$

This is LMARS written in m instead of $\mathbf{u}$: with $[\![p]\!] \approx P'[\![\Theta]\!] = (c^2/\theta)[\![\Theta]\!]$, $m^* = \{\{m_n\}\} - [\![p]\!]/(2c)$ and $p^* = P'\Theta^* = \{\{p\}\} - \frac{1}{2}c\,[\![m_n]\!]$, i.e. the LMARS interface velocity and pressure $w^* = \{\{u_n\}\} -$

$[\![p]\!]/(2\rho c)$, $p^* = \{\{p\}\} - \frac{1}{2}\rho c[\![u_n]\!]$ with the density factor absorbed into the momentum. No density upwinding ρ_{up} and no $1/\rho$ appear: m^* is linear in $(m, \Theta)_z$.

The interface fluxes (per unit contravariant normal, outward from L):

$$\hat{f}_\rho = m^*, \qquad \hat{f}_\Theta = \theta_{\mathrm{up}}\, m^*, \qquad \hat{f}_m^{\mathrm{adv}} = m_y^*\, \mathbf{u}_{\mathrm{up},y} \quad \text{(slow)},$$

θ_{up}, $\mathbf{u}_{\mathrm{up}}$ the values of the upwind side, the side chosen by the sign of $m^*(y)$. In strong form the surface term at a face node of element L is

$$\frac{s}{J\,w}\Big(\hat{f} - f_L \cdot \mathbf{n}\Big), \qquad f_{\rho,L}\cdot\mathbf{n} = m_{n,L}, \quad f_{\Theta,L}\cdot\mathbf{n} = \theta_L m_{n,L},$$

and for the non-conservative pressure term the lifted jump to the interface value,

$$\frac{s}{J\,w}\, P'(\Theta_{y,L})\,\big(\Theta^* - \Theta_{z,L}\big)\,\mathbf{n}.$$

Frozen face coefficients. $\theta_f = \{\{\theta_y\}\}$, $c_f = \{\{c_y\}\}$ (or max), the upwind side and θ_{up} from y. With z evolving, m^* and Θ^* are linear in z; the face part of the fast operator needs $c_f/(2\theta_f)$, $\theta_f/(2c_f)$, θ_{up} and the nodal P'_L, the slow part $m^*(y)\,\mathbf{u}_{\mathrm{up}}(y)$.

Contact discontinuities. Because $p = P(\Theta)$, a jump in Θ is a jump in p. A contact (constant p, jump in θ and ρ) has $[\![\Theta]\!] = 0$ and gets no acoustic dissipation in m^*; θ is upwinded in $\hat{f}_\Theta$, ρ is carried by the central $\{\{m_n\}\}$.

Walls. Mirror state $m_{n,R} = -m_{n,L}$, $\Theta_R = \Theta_L$: $m^* = 0$ and $\Theta^* = \Theta_L + \theta_f m_{n,L}/c_f$, so the pressure surface term is $P'_L(\Theta^* - \Theta_L) = c\, m_{n,L}$ (with $P'\theta = c^2$), the LMARS wall pressure $p + \rho c\, u_n$.

3.4 Hydrostatic balance

No background state is subtracted. A hydrostatic state given at the nodes, continuous across faces, has no jumps ($\Theta^* = \Theta$, $m^* = 0$ at interior faces and walls), so the discrete momentum residual is the volume term alone, $r_a = P'(\Theta_a)\,(\mathcal{G}\Theta)_a + \rho_a g\,\mathbf{e}_z$, i.e. the pointwise error of the degree-p differentiation of $\Theta(z)$, of order Δz^p per element (spectral in p). An initial state in exact discrete balance is obtained by solving $P'(\Theta_a)(\mathcal{G}\Theta)_a = -\rho_a g$ for ρ_a at given Θ (nodewise, since the operator acts on Θ only), which needs no change of the scheme. Section 4 gives a variant that is exactly balanced for the isothermal and the constant-θ atmosphere, still without subtraction.

4 Well-balanced variant

Artiano, Knoth, Spichtinger, Ranocha (*Structure-preserving high-order methods for the compressible Euler equations in potential temperature formulation for atmospheric flows*, arXiv:2509.10311) obtain well-balanced DGSEM schemes without any background state: gravity $-\rho\nabla\phi$ ($\phi = gz$ the geopotential, a given function) is discretized as a non-conservative two-point term $\bar{\rho}_{ab}[\![\phi]\!]_{ab}$, the pressure by the two-point flux $\{\{p\}\}$, and the mean $\bar{\rho}$ is chosen such that $\bar{\rho}[\![\phi]\!] = -[\![p]\!]$ holds exactly for the hydrostatic state:

$$\text{isothermal:}\quad \bar{\rho} = \{\{\rho\}\}_{\log} = \frac{[\![\rho]\!]}{[\![\log\rho]\!]}, \qquad \text{constant } \theta:\quad \bar{\rho} = \{\{\rho\}\}_\gamma = \frac{\gamma-1}{\gamma}\,\frac{[\![\rho^\gamma]\!]}{[\![\rho^{\gamma-1}]\!]}.$$

(Isothermal: $p = RT\rho$, $[\![\log\rho]\!] = -[\![\phi]\!]/(RT)$, so $\{\{\rho\}\}_{\log}[\![\phi]\!] = -RT[\![\rho]\!] = -[\![p]\!]$. Constant θ: $p = K\theta^\gamma\rho^\gamma$ and $\frac{\gamma}{\gamma-1}K\theta^\gamma\rho^{\gamma-1} + \phi = \text{const}$, so $\{\{\rho\}\}_\gamma[\![\phi]\!] = -K\theta^\gamma[\![\rho^\gamma]\!] = -[\![p]\!]$.) Since ϕ is continuous, gravity has no surface term.

Product form with two-point coefficients. The product form fits this exactly if $\mathrm{d}p/\mathrm{d}\Theta$ is frozen as a *two-point secant* where it appears. Along a ξ line (pair $a = (i, j)$, $b = (k, j)$, $\{\{J\xi\}\}_{ab} = \frac{1}{2}\big((J\xi_x, J\xi_z)_a + (J\xi_x, J\xi_z)_b\big)$; η lines likewise):

$$(\text{pressure})_a = \frac{1}{J_a}\sum_k D_{ik}\, P'_{ab}\, [\![\Theta]\!]_{ab}\, \{\{J\xi\}\}_{ab}, \qquad P'_{ab} = \frac{[\![p]\!]_{ab}}{[\![\Theta]\!]_{ab}} \quad \big(\to P'(\Theta) \text{ for } \Theta_a = \Theta_b\big),$$

$$(\text{gravity})_a = \frac{1}{J_a}\sum_k D_{ik}\, \bar\rho_{ab}\, [\![\phi]\!]_{ab}\, \{\{J\xi\}\}_{ab}.$$

With the secant, $P'_{ab}[\![\Theta]\!]_{ab} = [\![p]\!]_{ab}$, and since $2\{\{p\}\}_{ab} = 2p_a + [\![p]\!]_{ab}$ and $\sum_{\xi,\eta}\sum_k D_{ik}\{\{J\xi\}\}_{ab} = 0$ (discrete metric identities), the pressure term is identical to the conservative flux differencing $\frac{1}{J_a}\sum_k 2D_{ik}\{\{p\}\}_{ab}\{\{J\xi\}\}_{ab}$: the product form loses nothing. For the hydrostatic state every pair vanishes separately, $P'_{ab}[\![\Theta]\!]_{ab} + \bar\rho_{ab}[\![\phi]\!]_{ab} = [\![p]\!]_{ab} + \bar\rho_{ab}[\![\phi]\!]_{ab} = 0$, so the volume residual is zero for any degree and mesh. At the faces the state is continuous and at rest: $[\![\Theta]\!] = 0$, $m^* = 0$, $\Theta^* = \Theta_L$, and the pressure surface term vanishes as well. With the face secant $P'_{L*} = \big(P(\Theta^*) - P(\Theta_L)\big)/(\Theta^* - \Theta_L)$ the surface term $P'_{L*}(\Theta^* - \Theta_L) = p(\Theta^*) - p_L$ is the conservative one, too.

Frozen form. Freeze the coefficients where they appear, from y:

$$P'_{ab}(y) = \frac{[\![p_y]\!]_{ab}}{[\![\Theta_y]\!]_{ab}}, \qquad R_{ab}(y) = \frac{\bar\rho_{ab}(y)}{\{\{\rho_y\}\}_{ab}}, \qquad P'_{L*}(y),$$

and take $[\![\Theta_z]\!]_{ab}$, $\Theta^*_z - \Theta_{z,L}$ and $R_{ab}\{\{\rho_z\}\}_{ab}[\![\phi]\!]_{ab}$ live. The fast part stays linear in z, $F(q, q) = N(q)$ holds, and at $y = z = q$ hydrostatic the scheme is exactly balanced. This is the same secant as in the FV frozen split (`FV_FROZEN_PRESSURE = :derivative`); the frozen gravity ratio R is the one of the present conservative implementation.

Other stratifications. For a general hydrostatic profile (e.g. constant Brunt–Väisälä frequency, the gravity-wave tests) neither mean is exact; the residual is then of the truncation order of the mean, $\bar\rho - \rho^{\text{exact}}_{ab} = O(\Delta z^2)$ per pair. Choosing $\{\{\rho\}\}_\gamma$ (exact for constant θ) is the natural default for the troposphere; a profile-adapted mean would have to be a mean of ρ alone (horizontal pairs have $[\![\phi]\!] = 0$ and must not see it).

5 Summary of the split

term	frozen from y	live from z
volume mass flux	–	$\{\{m\}\}$
volume Θ flux	$\{\{\theta\}\}$	$\{\{m\}\}$
pressure (volume and surface)	nodal $P' = \gamma p/\Theta$ (well-balanced: secants P'_{ab}, P'_{L*})	$\mathcal{G}\Theta$ ($[\![\Theta]\!]_{ab}$), Θ^*
gravity	– (well-balanced: $R_{ab} = \bar\rho_{ab}/\{\{\rho\}\}_{ab}$)	ρ ($\{\{\rho\}\}_{ab}$)
face m^*, Θ^*	c_f/θ_f, θ_f/c_f	$\{\{m_n\}\}, [\![\Theta]\!], \{\{\Theta\}\}, [\![m_n$
face Θ flux	θ_{up}, upwind side	m^*
momentum advection (slow)	everything	–

6 Properties

1. $F(q, q) = N(q)$ by construction: N *is* the product-form DGSEM with these fluxes.

2. The fast part is linear in z. A HEVI fast loop needs one linear solve per column with the frozen vertical coefficients (no Newton iteration for $p(\Theta)$, unlike the conservative form, where the nodal $p(\Theta_z)$ is nonlinear). The coefficients of a stage are the b_{ij}-weighted means of θ, P', c_f/θ_f over the stage values.

3. Mass and Θ are conserved; momentum is not (non-conservative pressure).

4. Linear acoustics: for constant frozen coefficients the volume part is skew-adjoint in the energy $\frac{1}{2}\sum wJ\big(|m|^2 + (c^2/\theta^2)\Theta^2\big)$ (SBP), and the exact Riemann solver (4) adds the dissipation $-\sum_{\text{faces}} s\, w\, \frac{c}{2}\big(\llbracket m_n \rrbracket^2 + \frac{c^2}{\theta^2}\llbracket \Theta \rrbracket^2\big) \le 0$ (upwind flux of the symmetrized variables $(m_n,\, c\,\Theta/\theta)$). Variable coefficients add terms of the size of their variation, as usual.

5. Only θ and P' (and the face sound speed) are frozen, the "coefficients" of the fast operator; no part of the stiff response sits in the frozen argument. This is the rule of thumb found for the FV frozen split (freezing p/Θ instead of $\mathrm{d}p/\mathrm{d}\Theta$ was unstable).

7 Relation to the current implementation and open points

- `src/DGSEM/Euler2D/frozen.jl` splits the *conservative* DGSEM (pressure $\{p\}$ in the two-point momentum flux, gravity with a logarithmic mean of ρ, LMARS in the velocity): there the frozen coefficients are A/ρ_a, A/ρ_b, ρ_{up}/ρ, the ratio of the logarithmic to the arithmetic mean, and the fast part keeps the nonlinear nodal $p(\Theta_z)$. The product form removes all $1/\rho$ factors and makes the fast part linear.

- MIS stability. In both forms the Θ flux is $\theta_y m_z$: linearized about a wind $\bar{u}$, its fast part is $\bar{\theta}\,\delta m = \bar{\theta}(\bar{u}\,\delta\rho + \bar{\rho}\,\delta u)$ and its slow part $\bar{m}\,\delta\theta$. Density is advected in the fast part, Θ advection is split. On the column Fourier blocks this coupling limits the stability of MIS methods (`MethodDesign/app_stability.jl`; methods designed for it: `MIS_DG53`). The analysis has to be repeated for the product form.

- To do: implementation (`frozen_ncoef`, `frozen_data!`, `frozen_fast!` with the product-form N), test $F(q,q) = N(q)$, the frozen HEVI column solve, gravity-wave and terrain tests, and a comparison of the product form with the conservative DGSEM.